速成教育
SPEED UP EDUCATION

APSC160

Class 1



个人简介



Vincent (VC)

- CS Specialist – Software Engineering
- Master in Game Development
- 10年 受教经验
- 曾任 学校助教 及 助教培训导师, Lecture讲师 及 考试出题者
- steam 平台上 知名游戏的开发者
- 美股 / 加密货币 Day Trader
- 喜欢旅游 ☺️

课程分值组成 - Vantage

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- Participation	2%	
- Assignments (10)	10%	
- Labs (6)	18%	
- Term Test 1	12.5%	
- Term Test 2	12.5%	
- Final Exam	45%	Final 必须拿50%

课程分值组成

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- Worksheets	15%	
- Labs (best 8 of 9)	25%	
- Term Test 1	15%	
- Term Test 2	15%	
- Final Exam	30%	Final 必须拿50%

WTF is going on?

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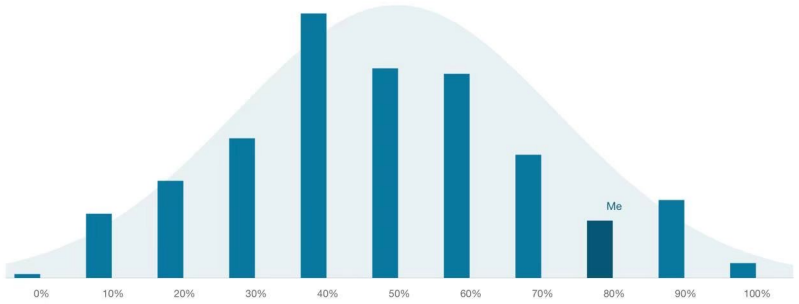
Term Test 1

Class scores distribution Hide

Percentages | Points

Total | Q1 | Q2 | Q3 | Q4 | MC p1

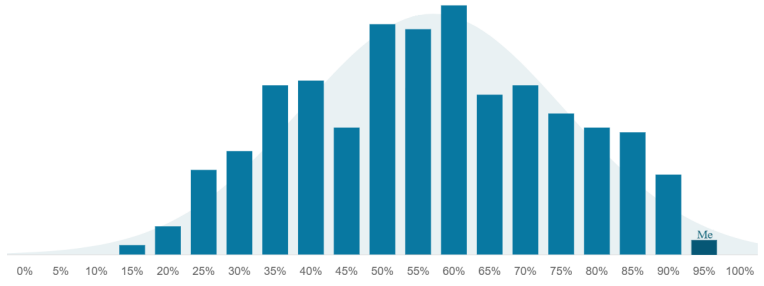
Students: 918 Mean: 49.1 Median: 47.1 Std. Dev: 21.3



Percentages | Points

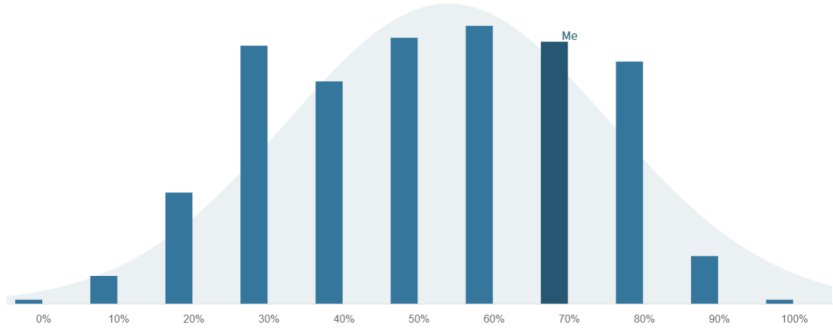
Total | Q1 pg 1 | Q1 pg 2 | Q2 | Q3 a) | Q3 b) | Q3 c) | Q3 d) | Q4 | Q5 | Q6 | Q7

Students: 631 Mean: 56.5 Median: 56.9 Std. Dev: 18.2



Total | Q1i | Q1ii | Q1iii | Q2i | Q2ii | Q2iii | Q3i | Q3ii | Q4i | Q4ii | Q4iii | MC p1

Students: 434 Mean: 53.2 Median: 54.5 Std. Dev: 20.3



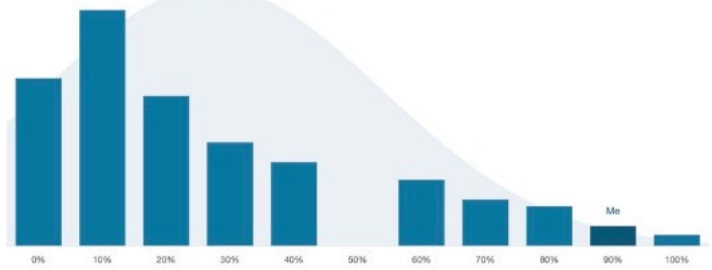
Term Test 1

Class scores distribution Hide

Percentages | Points

Total | Q1:IO | Q1:Loops | Q2:IO | Q2:If | Q2:Lists | Q3:Lists | Q3:Func | Q3:Doc | Q3:Test

Students: 419 Mean: 28 Median: 22.2 Std. Dev: 25



Make Up Test

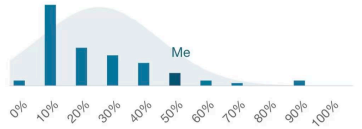
Class scores distribution

Hide

Percentages | Points

Total | Q1i | Q1ii | Q1iii | Q2 | Q3 | Q4i | Q4ii | Q4iii | MC p1

Students: 80 Mean: 24.6 Median: 20.6 Std. Dev: 18.1



Q1 | Q2i | Q2ii | Q3i | Q3ii | Q3iii | Q4i | Q4ii | MC

Students: 651 Mean: 44.3 Median: 40 Std. Dev: 19.8



课程难点

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1. **知其然，亦需知其所以然** (知晓运行原理, 代码必须是最简短且最有效率的)

2. **所有考试 都是 手写代码 没有电脑可用！ 基础一定要扎实, 绝不能临时抱佛脚**

   Final Exam 若没考过就会 auto fail

3. **Assignments & Labs 难度不简单**

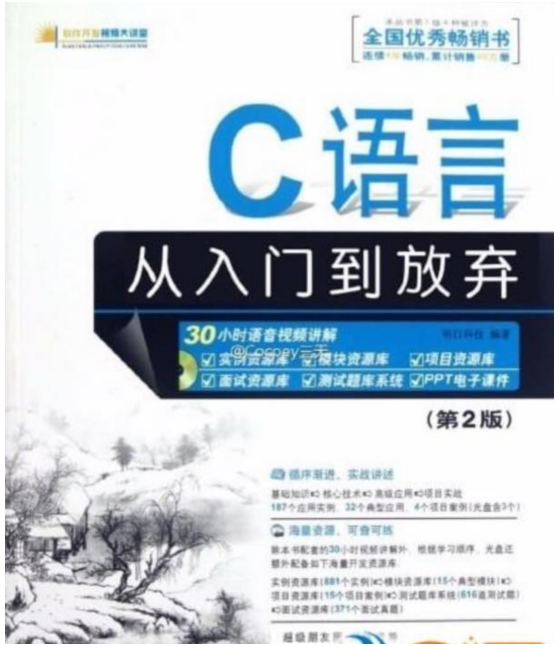
   有一套人工智能系统叫做 MOSS 的 查重软件!

   切勿互相讨论或分享代码！ ！ ！

   切勿互相讨论或分享代码！ ！ ！

   切勿互相讨论或分享代码！ ！ ！ (重要的事情说3遍)

   大学对抄袭作弊的检查/惩罚都非常严格，第一次扣分，第二次退学



我的课绝对不会给你任何作业/考试的答案, 我一定会让你自己有能力做出来!!

作业拿满分

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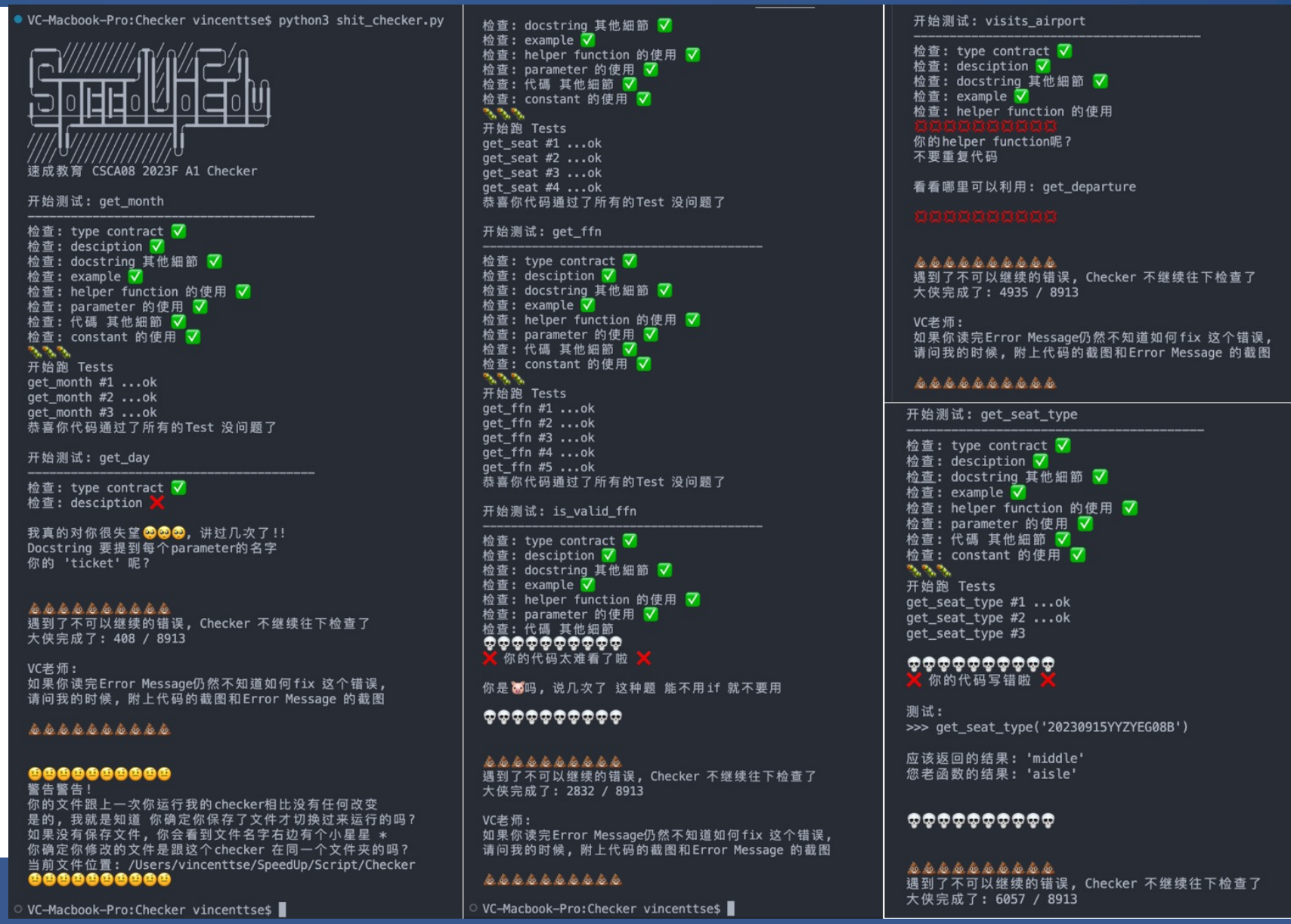
VC 的人工智障 AI Checker

每个 Lab,

都会提供专门的 **Checker**,

只要你过了 作业肯定 **满分**

且100% 是你自己写出来的!



学员好评反馈&高分成绩

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我的辅导班cs老师Vincent （vc），讲课很透彻 Mario上完课我一脸懵。vc再讲的时候 感觉突然好简单。课上轻松诙谐 人幽默风趣。深夜上课的时候 我昏昏欲睡 一句 len（12）给我点清醒了。 108 110 111 148 165都挺顶的。 台湾大gg 人也很帅



vincent老师！
有老师讲解我 tt1大题全部都满分做出来了
平常讲课的时候也是很搞笑的不会无聊
讲assignment也很用心保证我们是真的学会了



快来报 vc 的班 超有耐心的老师！！！跟着 vc 学到了很多！108保分大神



Cs大神 vc 老师课堂风趣幽默课后认真负责 有问必答 24小时在线 绝绝子 lecture上听不懂的vc都讲懂了 朋友们快



VC我的超人
代码的神 Python吴彦祖 德运龙字科V龙C



vc老师的csc108真的超级有用！
考前的review干货满满！对考试的预习很有帮助
教的也很有耐心，每次不仅有cover到学校的教材还有额外的练习题可以practice！
每次有问题老师都会回复的很及时

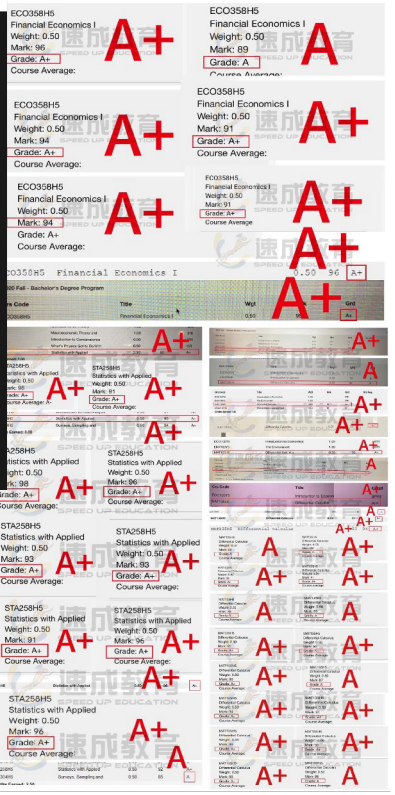


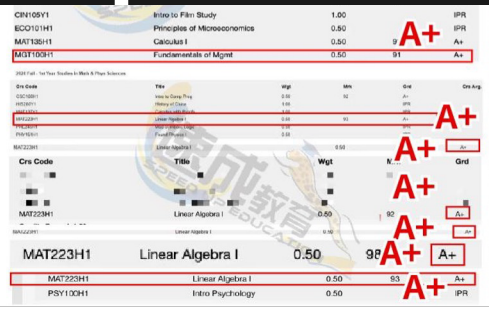
Vincent 老师，一个集搞笑与才华于一身的代码大神，课堂氛围幽默风趣，课后认真负责，有什么问题总是第一时间出现，总之就是爱了爱了～

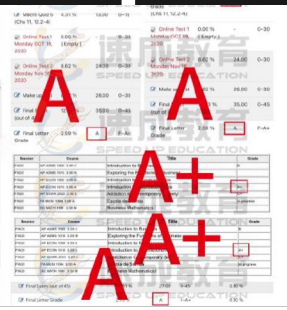


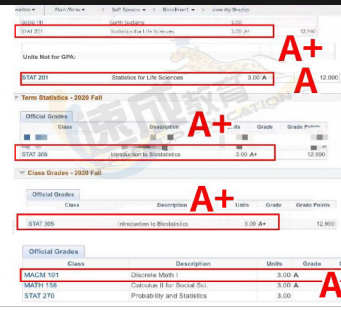
vc的课很有意思 所有知识点都讲的很详细并且每个代码块都会提供多种思路 让学生学到真正的技巧 授人以鱼不如授人以渔！ 而且课后的答疑也非常耐心 让学生把错综复杂的代码一点一点的有条不紊的连接起来！强烈推荐











VSCode 实际操作例子

C 语言 类型

Integer 整数 (*int*)

e.g. **1** **5**

Float 浮点数 (必须有小数点) (*float*)

e.g. **1.0** **3.14**

Char 字元 (*char*)

e.g. **'A'** **'b'**

Ascii Table	
A -> 65	a -> 97
B -> 66	b -> 98
...	...

今天这节课 就先当 `int` 和 `char` 是一样的...
实至上是有区别的, 其背后原理涉及到了
二进制的转换和存储.... 下周讲 👍

String 字符串(*char*, char[]*)

e.g. **"Hello"** **"APSC160"**

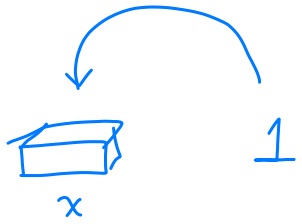
要充分理解 String 背后的运行原理 且在考试拿高分...
需要涉及到 ***Pointer***, ***Array*** 和 ***Memory Model***.
我需要讲2个小时 我们下周见!

VSCode 实际操作例子

Variable 变量

assign

variable = expression



Example 1

```
int x = 1;
printf(x);
printf("%d", x);
printf("x")
printf("%d", "x");
printf("%s", "x");
printf("%d", y);
```

```
> 1
  x
  x
NameErorr: 'y' is not defined
```

pen_island = ...

penIsland = ...

```
int a = 100;
int b = a;

a = 10;

printf("%d", a);
printf("%d", b);
```

10 / 100 / a

100 / b

10

100

Naming Convention

- × Cannot start with a number.
- × Cannot contain special symbols.
- × Cannot contain white spaces.

Example 2

```
int x = 1;
int y = 2;
x = y;
printf("%d", x);
printf("%d", y);
```

```
> 2
  2
```

Example 3

```
int x = 1;
int y = 2;

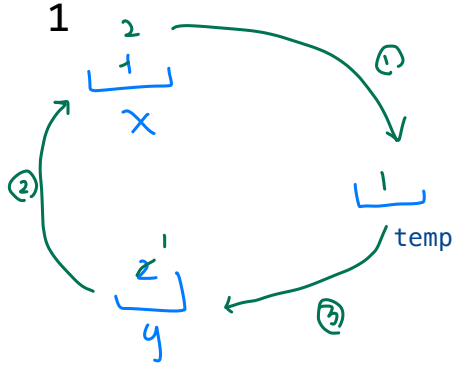
int temp = x;

x = y;

y = temp;

printf("%d", x);
printf("%d", y);
```

```
> 2
  1
```



Question 1:

What is the result of running the following code:

```
int main() {
  int a;
  printf("%d", a);
  a = 10;
}
```

a) 10

b) I dont know

c) The program runs but produces weird/incorrect output

d) The compiler throws out an error when your code is running

e) None of the above

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Question 2:

What is the result of running the following code:

```
int main() {
  int a = 10;
  printf("%d", a / 0);
}
```

a) 0

b) I dont know

c) The program runs but produces weird/incorrect output

d) The compiler throws out an error when your code is running

e) None of the above

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Question 3:

What is the result of running the following code:

```
int main() {
  int a = 10;
  printf("%d", b);
}
```

a) 0

b) I dont know

c) The program runs but produces weird/incorrect output

d) The compiler throws out an error when your code is running

e) None of the above

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Question 4:

Assume variable *x* and *y* has been defined.
One of the following options must be true after we execute the code `y = (char) x;`

a) *x* will be of type *char*

b) *y* will have a value equivalent to the *char* value of *x*

c) *y* will be of type *char*

d) *y* and *x* must have the same value

e) None of the above is certain to be true

VSCode 实际操作例子

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以下所有题目 请勿用 今天这份讲义以外的知识点 包括 if 语句, 循环, ... 等等

Assume the following functions exist to find the maximum and minimum of two values (a and b). The code has been omitted.

```
int max(int a, int b) {...}
int min(int a, int b) {...}
```

Example 1:

幼兒園春遊問題: 输入人数, 计算需要多少辆车, 一辆车可以坐40人 (至少10种不同写法)

```
#include <math.h>
int main() {
    int n;  printf("请输入人数: ");  scanf("%d", &n);

    printf("%d\n", ceil(n / 40));                    方法1

    printf("%d\n", n / 40 + min(1, n % 40));         方法2

    printf("%d\n", (n + 39) / 40);                   方法3

    printf("%d\n", ((n - 1) / 40) + 1);              方法4
 }
```

以下所有题目 请勿用 今天这份讲义以外的知识点 包括 if 语句, 循环, ... 等等

Example 2:

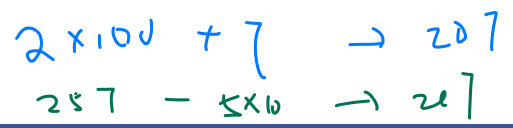
三位数字重组

输入任意的3位数字，返回一个新的数字，把之前的数字的十位数字变成0 （至少5种不同写法）

```
int main() {
    int n;  printf("请输入3位数字: ");  scanf("%d", &n);

}
```





例子

```
> ./a.out
  请输入3位数字: 257
  答案是: 207

> ./a.out
  请输入3位数字: 123
  答案是: 103
```

以下所有题目 请勿用 今天这份讲义以外的知识点 包括 if 语句, 循环, ... 等等

Assume the following functions exist to find the maximum and minimum of two values. The code has been omitted.

```
int max(int a, int b) {...}
int min(int a, int b) {...}
```

Example 3:

寻找中间数

给你三个数字，返回中间的那一个 (至少10种不同写法)

```
int main() {
    int x, y, z;
    printf("请输入3个数字 分别用刚好一个空格 隔开: ");
    scanf("%d %d %d", &x, &y, &z);
```

例子

```
> ./a.out
请输入3个数字 分别用刚好一个空格 隔开: 10 5 2
中间数 median 是: 5

> ./a.out
请输入3个数字 分别用刚好一个空格 隔开: 10 30 20
中间数 median 是: 20
```

以下所有题目 请勿用 今天这份讲义以外的知识点 包括 if 语句, 循环, ... 等等

Assume the following functions exist to find the maximum and minimum of two values. The code has been omitted.

```
int max(int a, int b) {...}
int min(int a, int b) {...}
```

Example 4:

加拿大人所得税

给你的收入,计算个人所得税，前20,000块是15%的税，超过20,000的部分都是25%的税（当然，实际计算方式 复杂很多）

```
int main() {
    int income;
    printf("请输入你的年薪: ");
    scanf("%d", &income);
```

例子

```
> ./a.out
请输入你的年薪: 35000
你需要缴 $6750.00 的税
```

```
> ./a.out
请输入你的年薪: 19000
你需要缴 $2850.00 的税
```

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Python 其他类型

Boolean 布尔值 (*bool*)

其實，除了0以外的數字都是true

e.g. **true (1)** **false (0)**

Ascii Table	
A -> 65	a -> 97
B -> 66	b -> 98
...	...

A	B	A && B	A || B	! A
T	T	T	T	F
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

Basic Operation 基本操作

Value 是否相等

```
>>> 1 > 2
false
>>> 2 <= 3
true
>>> 3 == 3.0
true
>>> 3 == '3'
False
>>> 3 != 3.0
False
```

```
>>> 'b' > 'a'
>>> 'A' > 'a'
```

Lazy Evaluation 考试选择题 必考!!!

```
>>> 2 == 2.0 || 3 + "2" == 5
```

```
>>> 0 && 1 / 0 == 2
```

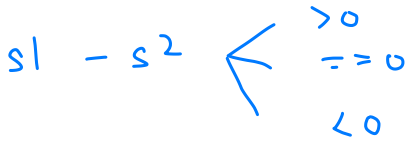
咱提前講『一些』下周的內容

Comparing Strings

```
#include <stdbool.h>
#include <stdio.h>
#include <string.h>
int main() {
    char s1[10] = "apple";
    char s2[10] = "app1e";
    int is_same = strcmp(s1, s2);
    printf("%d\n", is_same);
}
```

```
int strcmp(const char *s1, const char *s2);
```

Compares the two strings s1 and s2.
It returns an integer less than, equal to, or greater than zero if s1 is found respectively, to be less than, to match, or to greater than s2.



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2014年 Midterm 1 考试原题：

write a program to decide if a student should be accepted into a course, given the following rules:

- Anyone whose program code is CSC should be allowed into the course
- Students whose program code is MAT or STA require a GPA of 3.0 or higher to enter the course
- Any student with a GPA above 3.5 is allowed in
- Anyone named Vincent should be allowed into the course

You may **not** use any if statements, loops or any other elements we have not covered.

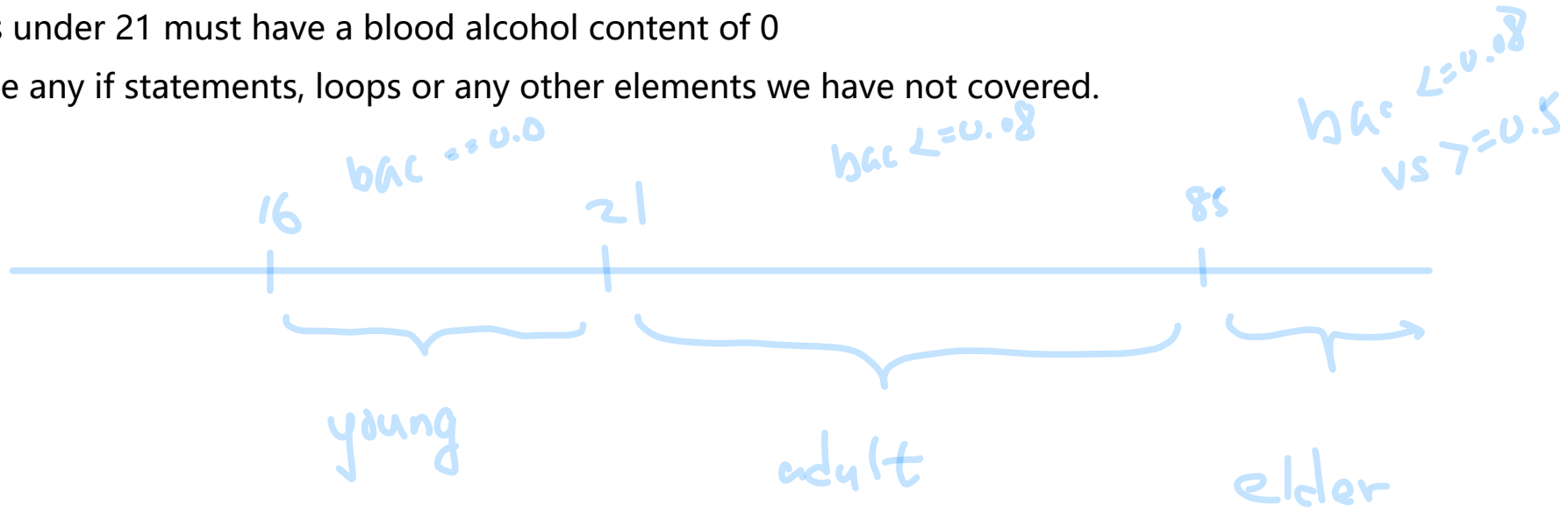
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2015年 Midterm 1 考试原题：

write a program to decide if a person is able to legally drive in Canada, given the following rules:

- All drivers must be 16 years of age or older
- All drivers over 85 must have a vision score of 0.5 or better
- All drivers must have a blood alcohol content of 0.08 or lower
- Drivers under 21 must have a blood alcohol content of 0

You may **not** use any if statements, loops or any other elements we have not covered.



UBC 2026秋季学期 精品周课班

新生专属

限时

超低折扣

扫码咨询详情

整学期 28+小时课时

- 全职硕博资深教职团队
- 每周一课紧贴学校内容
- 课后1V1答疑随时在线
- 占分作业小测满分讲解

海量真题 考前演练 · 视频课程 随时回看

全程教学质量保障 | 不满意可随时申请退